



Department of Computer Science and Engineering
National Institute of Technology, Rourkela
End Semester Examination (Spring 2025 - 26)
MTech 1st Semester
Sub: Foundations of Computer Security (CS - 6301)
Time: 3 Hrs; FM: 50

Answer **ALL** questions. Figure in the right hand margin indicates the marks.

1. (a) Find the result of $21^{24} \bmod 8$ using the square-and-multiply method. [5] [CO3, BL2]
(b) Solve $x^2 \equiv 12 \bmod 17$ using quadratic residue. [5] [CO2, BL3]
2. (a) Show the first eight words of the key expansion in AES for a 192-bit key of all ones. [The key consists of all 1's] [5] [CO4, BL3]
(b) What are the advantages and disadvantages of AES over DES with respect to the key size? [5] [CO4, BL3]
3. (a) Compare the permutations in DES and AES. Why do we need expansion and compression in DES, but not in AES? [5] [CO4, BL3]
(b) Use the extended Euclidean algorithm to find the inverse of $x^4 + x^3 + 1$ in $GF(2^5)$ using the modulus $x^5 + x^2 + 1$. [5] [CO3, BL3]
4. (a) Solve the following congruence equation using the Chinese Remainder Theorem:
$$x \equiv 2 \bmod 7$$
$$x \equiv 3 \bmod 9$$
[5][CO3, BL2]
(b) Solve $x^5 \equiv 11 \bmod 17$ using the properties of discrete logarithms. [5][CO2, BL3]
5. Answer the following questions: [5x2 = 10]
 - a. Find the n-bit word represented by the polynomial x^7 in $GF(2^8)$. [CO3, BL2]
 - b. Find the order of the group $G = \langle \mathbb{Z}_{19}^*, x \rangle$. [CO3, BL2]
 - c. What is the difference between SubBytes and SubWord in AES? [CO2, BL1]
 - d. Distinguish between diffusion and confusion. [CO3, BL1]
 - e. Find the value of $\phi(101)$. [CO4, BL2]